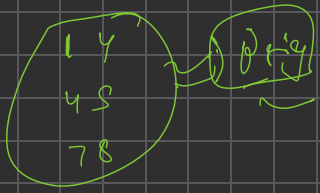
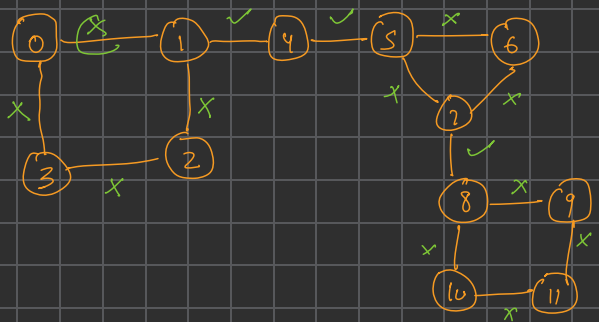
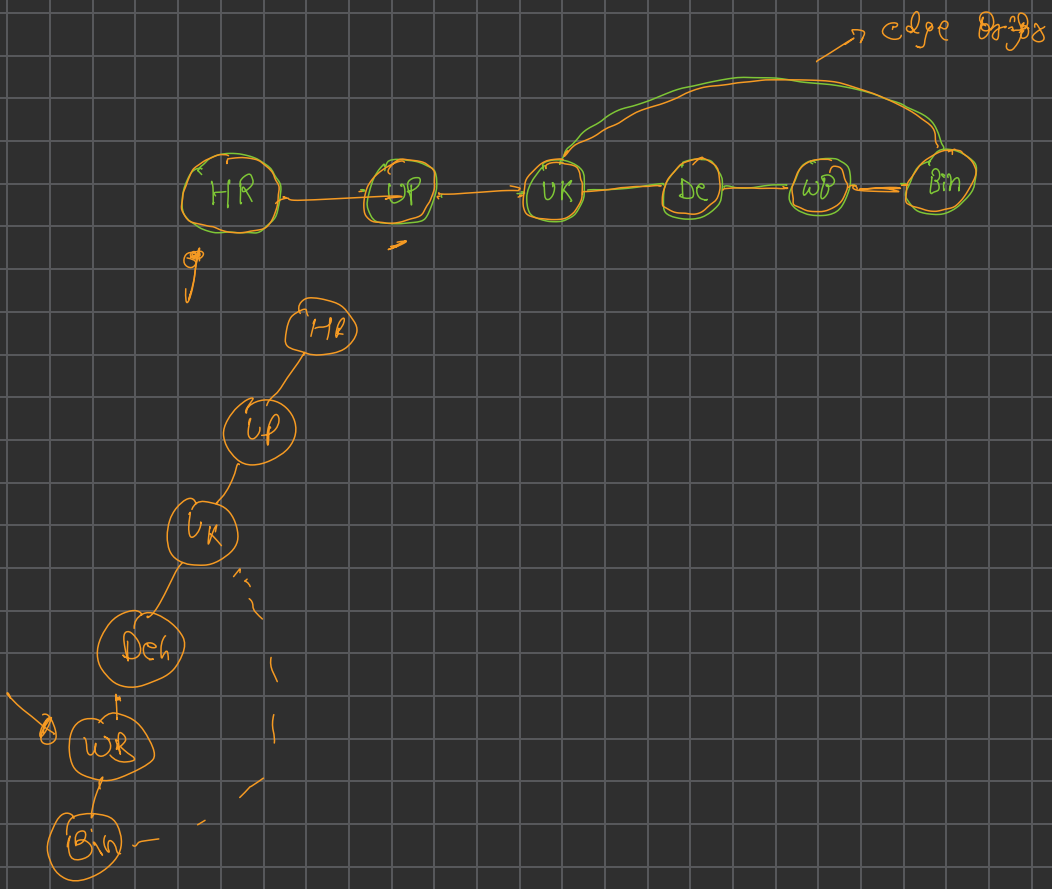
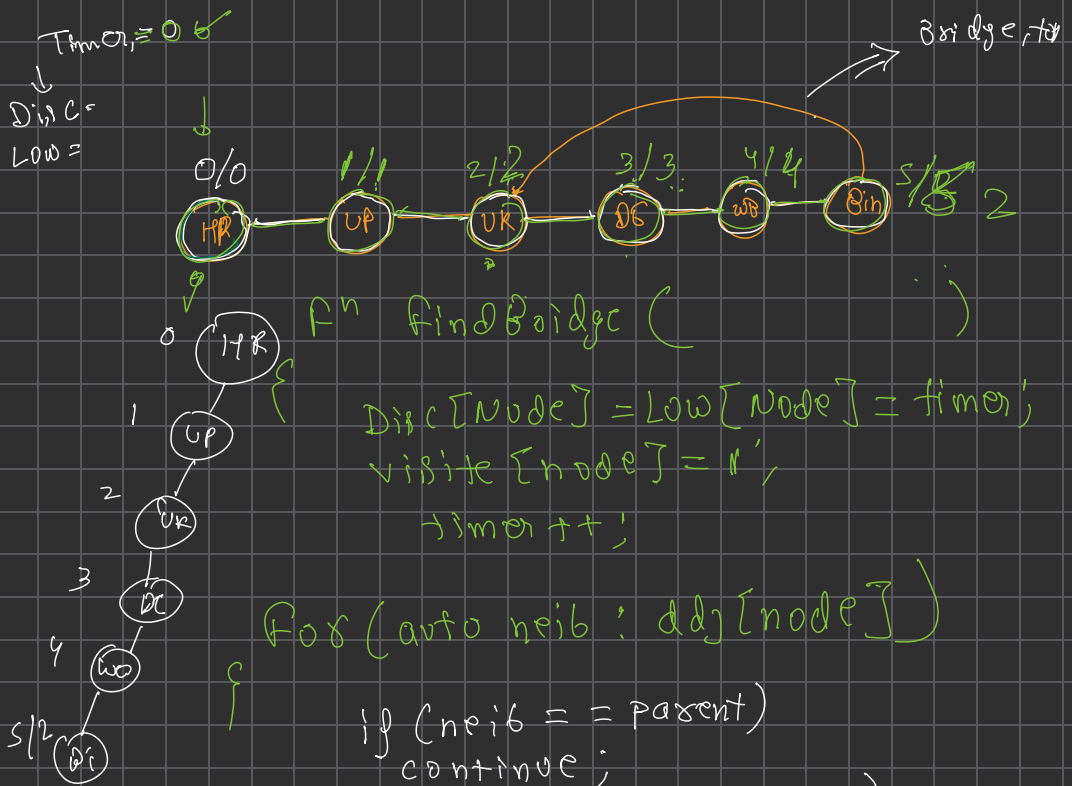
[Bridge in a graph]







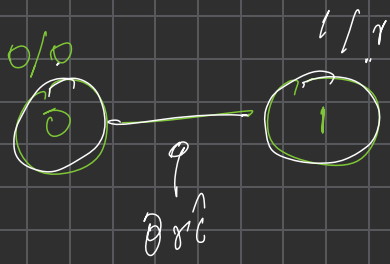
if disc[node] < low[neib]
bridge hai

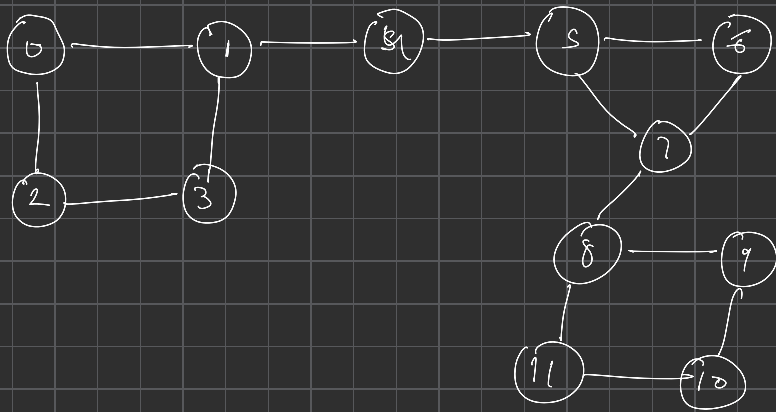


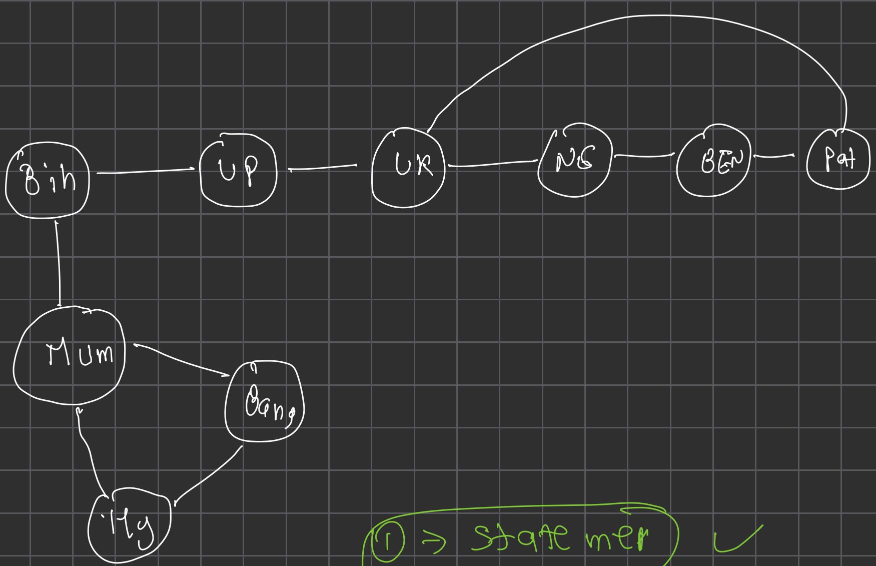

```
fn findBridge ( )
{
    Disc[Node] = Low[Node] = timer;
    visite[node] = 1;
    timer++;

for(auto neib : adj[node])
{
    if (neib == parent)
        continue;
    else if ( visited[neib])
    Low[node] = min [ Low[neib], Low[node]
    else
    {
        find bridge( )
        if ( disc[node] < Low[neib]
            bridge++;
    }

    Low[node] = min ( Low[node],
                      Low[neib] );
}
```


0/0
0
1
1/1
a
src





① ⇒ state mer ✓

↳ 1 se jada ci

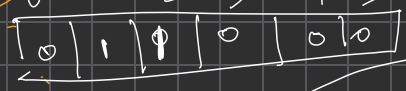
⇒ articulation

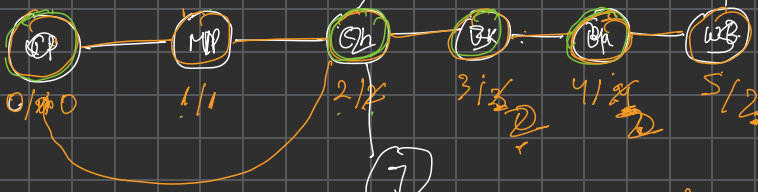
Edge case handle

array store?

child

0 1 2 3 4 5





Non start node

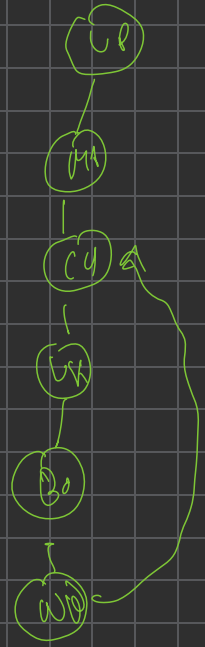
Articulation point

if (disc[node] <= low[neig])

{ I am articulation .

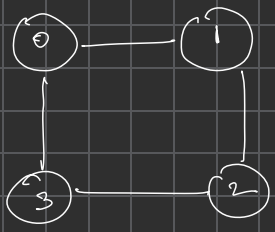
if (visited [neig]

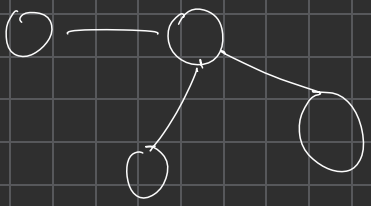
low [node] = min (low [node], disc [neig])



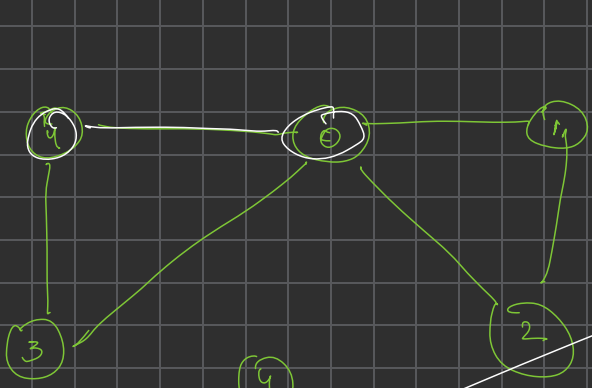
Hamiltonian

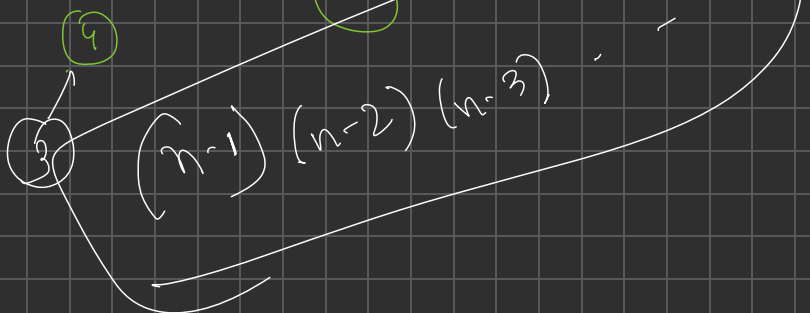
(1) node

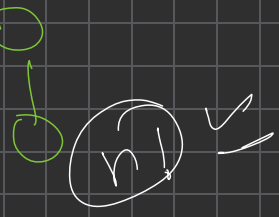




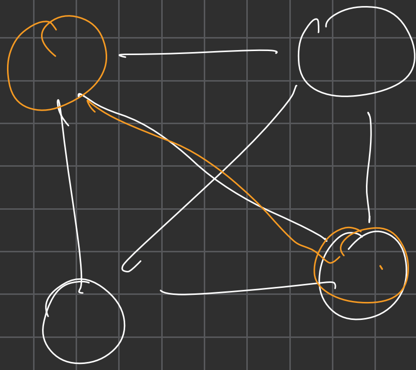
You have to traverse all the node exactly one time







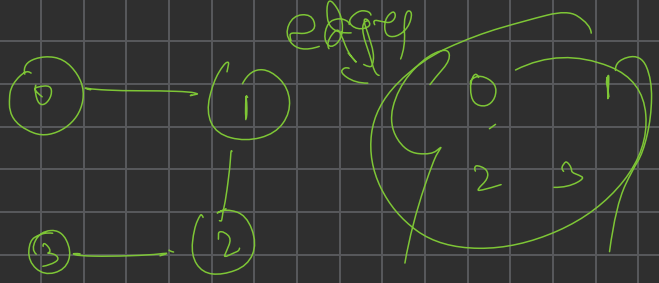
n :

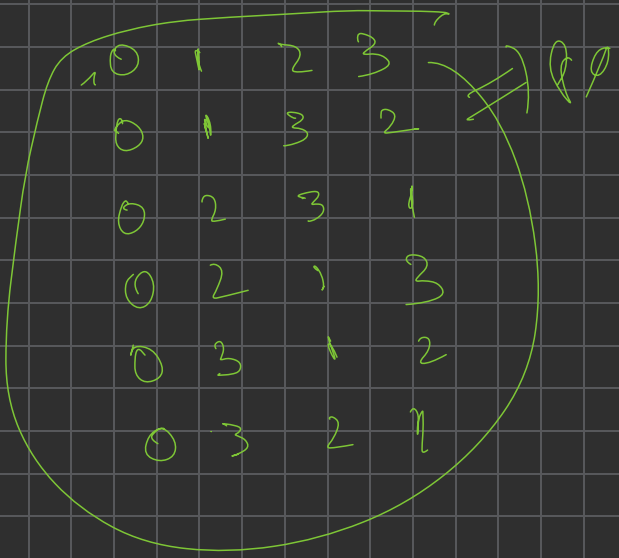


6 -

$\Rightarrow n! \Rightarrow$ Fact

$(n-1) \times (n-2) \times (n-3)$


edge
0
1
2
3
0
1
2
3


0 1 2 3
0 1 3 2
0 2 3 1
0 2 1 3
0 3 1 2
0 3 2 1